

Modern periodic Law : Modern periodic law invented by Dmitri Mendeleev. Mendeleev's Periodic Law — "The physical and chemical properties of elements are the periodic function of their atomic masses."

Some properties / key features of Modern periodic Table :

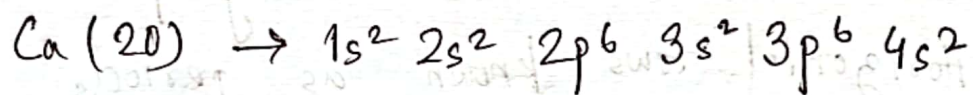
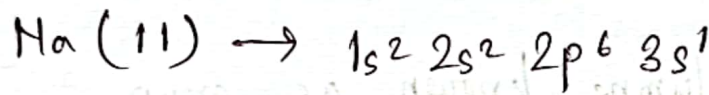
- 1) 18 vertical columns known as group
- 2) 7 Horizontal rows known as periods
- 3) Light metals → These are elements of periodic table of group 1 and 2
- 4) Heavy metals / Transition metals → These are elements of periodic table of group 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12
- 5) Non-metals → These are elements of periodic table of group 13, 14, 15, 16 and 17
- 6) Zero group / Noble gas → These are elements of periodic table of group 18.

Classification on elements based on electronic configuration,

1) s-block elements | In electronic configuration, the elements whose last electron goes to s orbital are called s-block elements.

The elements of group 1 (alkali metals) and group 2 (alkaline earth metals) belongs to the s-block elements.

Examp: |

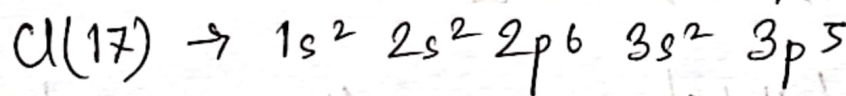
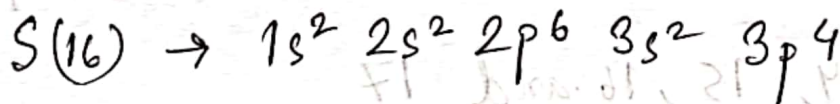


2) p-block elements | In electronic configurations

the elements whose last electron goes to p orbital are called p-block element.

The element of group -13 to group -18 belongs to the p-block elements.

Examp: |



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3) d-block elements: In electronic configuration, the elements whose last electron goes to d orbital are called d-block elements.

The elements of group 3 to group 12 belongs to the d-block elements.

Examp: | Cr (24) $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
Cu (29) $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

4) f-block elements: In electronic configuration, the elements whose last electron goes to f orbital are called f-block elements.

The elements of ~~group~~ lanthanides and Actinides series belongs to the f-block elements.

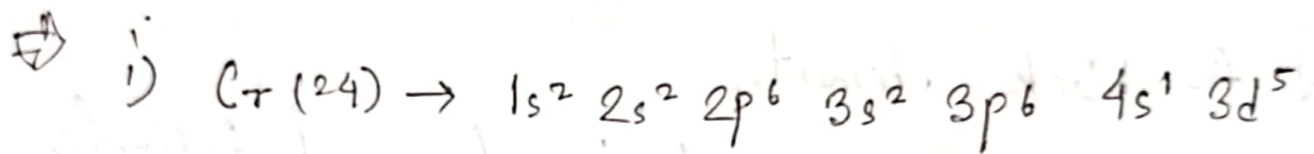
Examp: | La (57) $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$
 $5s^2 4d^{10} 5p^6 6s^2 4f^1$
Ce (58) $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$
 $5s^2 4d^{10} 5p^6 6s^2 4f^2$

The Rules of electronic configuration:

- ① Highest number of n is the period
- ② For group determination following formula will be used $(n-1)d + ns + np$

Ex. Write down electronic configuration and find out the period and group of the following atomic number:

- i) Cr₂₄ ii) Cu₂₉ iii) Kr₃₆ iv) Br₃₅

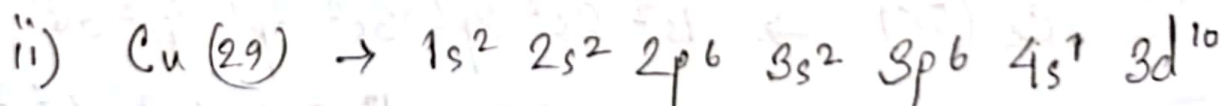


period = 4

Group = ~~(4-1)~~ $(n-1)d + ns + np$

$$= (4-1)d + 4s + 4p$$

$$= 3d + 4s + 4p = 5 + 1 + 0 = 6$$

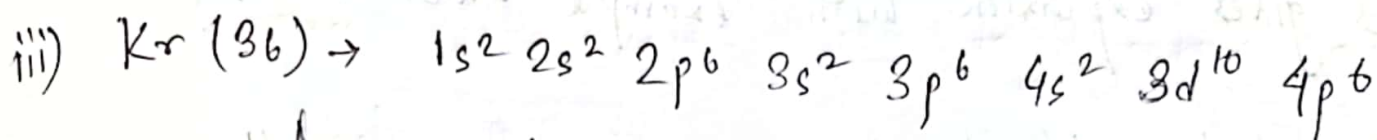


period = 4

Group = $(n-1)d + ns + np$

$$= (4-1)d + 4s + 4p = 3d + 4s + 4p$$

$$= 10 + 1 + 0 = 11$$



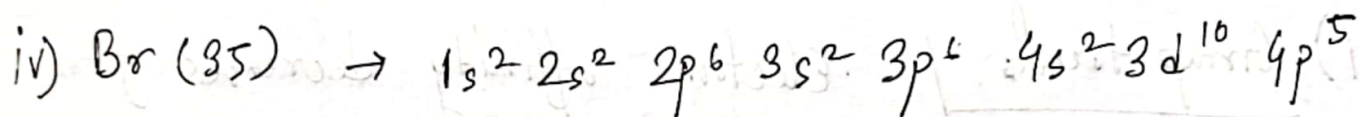
$$\text{period} = 4$$

$$\text{group} = (n-1)d + ns + np$$

$$= (4-1)d + 4s + 4p$$

$$= 3d + 4s + 4p$$

$$= 10 + 2 + 6 = 18$$



$$\text{period} = 4$$

$$\text{group} = (n-1)d + ns + np$$

$$= (4-1)d + 4s + 4p$$

$$= 3d + 4s + 4p$$

$$= 10 + 2 + 5 = 17$$

Periodicity: Periodicity refers to trends or recurring variations in element properties with increasing atomic number.

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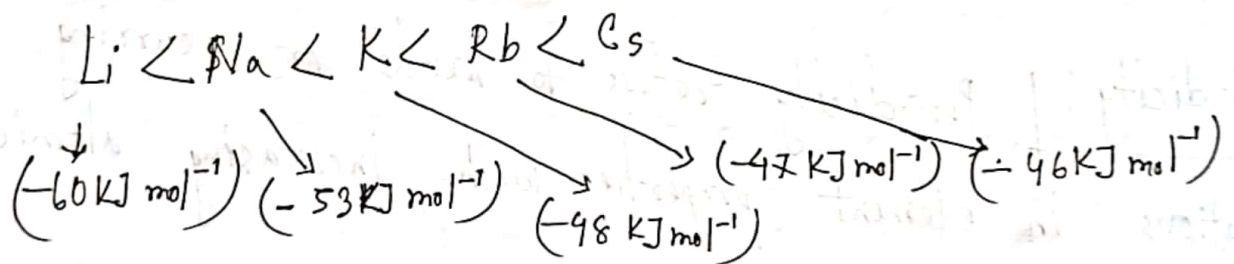
Q. How electron affinity differs in periodic table and gives explanation with example and values?

→ Electron Affinity: Electron affinity is the ability of an atom to attract and hold an extra electron.

The electron affinity differs in periodic table is given below:

1) Group trends: Electron affinity decreases from top to bottom of a group. This is because it's easier for small atoms with the nucleus closer to the outer electrons to gain another electron.

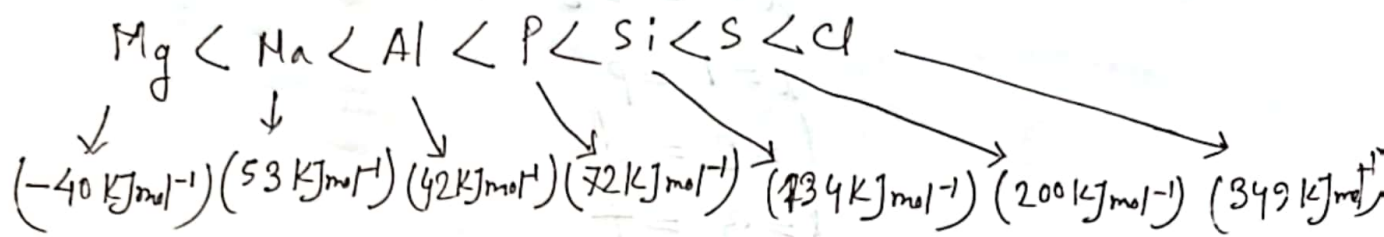
Examp: For Group 1:



2) Period trend: Electron affinity in a horizontal period increases from left to right. This is because the desire to gain an electron increases the closer

you get to fill the energy level.

Examp: 1 For Period 3:



Here, Mg & Na, P & Si have exception in electron affinity.

Q. Arrange the following orbitals according to order of higher energy

7p, 6d, 3d, 5d, 4f, 4s

⇒ Given that,

7s, 6d, 3d, 5d, 4f, 4s

$$7s \rightarrow (n+l) \rightarrow (7+0) \rightarrow 7$$

$$6d \rightarrow (n+l) \rightarrow (6+2) \rightarrow 8$$

$$3d \rightarrow (n+l) \rightarrow (3+2) \rightarrow 5$$

$$5d \rightarrow (n+l) \rightarrow (5+2) \rightarrow 7$$

$$4f \rightarrow (n+l) \rightarrow (4+3) \rightarrow 7$$

$$4s \rightarrow (n+l) \rightarrow (4+0) \rightarrow 4$$

$$\therefore 6d > 7s > 5d > 4f > 3d > 4s \quad \text{(Ans)}$$

Q. Why $1d/3f$ is not possible?

Ans. We know that,

$$l = 0 = s$$

$$l = 1 = p$$

$$l = 2 = d$$

$$l = 3 = f$$

If $l > n$ so electron configuration is not possible.

For $1d$:

$$n = 1 \quad \text{and} \quad l = 2$$

$\therefore l > n$ so $1d$ is not possible.

For $3f$:

$$n = 3 \quad \text{and} \quad l = 3$$

$\therefore l = n$ so $3f$ is not possible.

Q. Which of the following sets of quantum numbers are allowable or not allowable and why?

i) $n = 4$

$l = 1$

$m = +1$

$s = +1/2$

ii) $n = 2$

$l = 0$

$m = -2$

$s = -1/2$

iii) $n = 3$

$l = 3$

$m = 0$

$s = +1/2$

iv) $n = 4$

$l = 2$

$m = +2$

$s = -1/2$

$$v) \quad n=1 \quad l=0 \quad m=0 \quad s=+1/2$$

$$vi) \quad n=4 \quad l=3 \quad m=+4 \quad s=-1/2$$

$$\rightarrow \textcircled{i} \quad n=4 \quad l=1 \quad m=+1 \quad s=+1/2 \quad (\text{Allowable})$$

$$\textcircled{ii} \quad n=2 \quad l=0 \quad m=-2 \quad s=-1/2 \quad (\text{Not Allowable})$$

$\rightarrow (n=2 \text{ but } m \neq -2. \text{ If } n=2$
 $m = (-1, 0, 1))$

$$\textcircled{iii} \quad n=3 \quad l=3 \quad m=0 \quad s=+1/2 \quad (\text{Not Allowable})$$

$\rightarrow (n=3 \text{ but } l \neq 3. \text{ If we know}$
 $n > l \text{ and } l = n-1$
 $\therefore l = 3-1 = 2)$

$$\textcircled{iv} \quad n=4 \quad l=2 \quad m=+2 \quad s=-1/2 \quad (\text{Allowable})$$

$$\textcircled{v} \quad n=1 \quad l=0 \quad m=0 \quad s=+1/2 \quad (\text{Allowable})$$

$$\textcircled{vi} \quad n=4 \quad l=3 \quad m=+4 \quad s=-1/2 \quad (\text{Not Allowable})$$

$\rightarrow (n=4 \text{ but } m \neq +4, m = \pm l$
 $\therefore l=3 \text{ so } m = +3 \text{ and } n > m)$
 is not possible

Q. Suppose an electron is in 4f electrons. Find out the probable four quantum number values for it.

Given that,
 $n = 4$

So,

l	0	1	2	3
m	0	+1 0 -1	+2 +1 0 -1 -2	+3 +2 +1 0 -1 -2 -3
s	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$

Now for 4f

$\therefore n = 4$ and $l = 3$ //

m	+3	+2	+1	0	-1	-2	-3
s	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$	$\pm 1/2$

// (Ans)